

YANGFAN (KEN) ZHENG

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EXPERIENCE

Software Engineer

August 2024 – Present

Deutsche Bank

Cary, NC

- Owned the full development cycle of CASP, the bank's daily position reconciliation service — a multi-tiered, distributed Spring Boot (Java 17) batch system processing 100K+ positions nightly with multi-threaded batch processing and JDBC bulk persistence tuned for 10x faster writes; replaced a manual workflow and cut break-detection turnaround from days to same-day.
- Designed the service's object-oriented parsing layer for fixed-width broker files, with composable field-binding abstractions so new file formats ship without touching core pipeline code; wrote the layout specs and runbooks the team uses for onboarding and support.
- Cut account-balance query resolution from hours to under 30 seconds by shipping a Spring Boot microservice on GCP Vertex AI (Gemini) that translates natural-language questions into governed SQL, reducing manual data collection by 40%.
- Architected Backend-for-Frontend (BFF) microservices behind the client-facing TAS Dashboard's unified transaction view with caching and aggregation strategies that cut page-load and data-retrieval time by 40%.
- Diagnosed and fixed production defects end to end — including a silent file-corruption bug in a PGP/SFTP transfer service and a duplicate-insert race in a concurrent batch job — and mentored a junior engineer through the reconciliation codebase and test suite.

Software Engineering Intern

June 2023 – August 2023

Deutsche Bank

Cary, NC

- Reduced false-positive alerts by 25% in a payment-surveillance system scanning 5M+ daily transactions by building a PySpark/Impala data pipeline surfaced through a React/TypeScript UI.
- Achieved 90%+ code coverage and a 99% defect-free release for new account integrations by designing 50+ automated PyTest test cases.
- Delivered ESG intelligence to internal stakeholders by classifying 10,000+ news articles daily with GCP NLP API, served via a Flask/Plotly Dash application.

PROJECTS

ReconPulse (Trade Reconciliation Platform) | Java 17, Spring Boot 3, AWS, MongoDB, Docker/Kubernetes 2026 – Present

- Designed and built an event-driven trade reconciliation platform on AWS — ingesting EOD trade files from S3 and real-time trade events from SQS into a pluggable rules-based matching engine with configurable tolerances, detecting five break types and persisting results in MongoDB with idempotent, index-backed writes.
- Built an AI-assisted break triage service on Amazon Bedrock that clusters reconciliation breaks and generates root-cause hypotheses — with counterparty/PII redaction at prompt construction, strict JSON schema validation of LLM outputs, and deterministic fallback on failure.
- Practiced TDD end to end with a full delivery pipeline — 90%+ coverage via JUnit 5/Mockito and Testcontainers integration tests (LocalStack, MongoDB), GitHub Actions CI with coverage gates, and Docker/Kubernetes deployment with health probes and HPA autoscaling.

AI Content Studio (YouTube Channel) | Java 17, Spring Boot, Python, ComfyUI, ffmpeg

2026 – Present

- Built an autonomous distributed pipeline that discovers, scores, and publishes anime video compilations to YouTube end to end: AI vision models rank 2,900+ archived clips, and ffmpeg assembles final videos with auto-generated creator credits.
- Engineered for cost and reliability — a staged selection funnel runs expensive AI scoring on the fewest clips possible, per-clip failure isolation keeps one bad input from killing a run, and uploads execute as async REST jobs with resumable OAuth.
- Built a companion generative image pipeline (FLUX.1 on ComfyUI) for a fully synthetic on-channel persona with reproducible Python-defined workflows on a locally tuned AMD GPU.

TECHNICAL SKILLS

Languages: Java, Python, SQL, JavaScript/TypeScript, C/C++

Backend & Design: Spring Boot, Object-Oriented Design, multi-threading/concurrency, Hibernate/JPA, Kafka, REST APIs, Django, Flask, JUnit, Mockito, PyTest

Cloud & Infrastructure: AWS (S3, SQS, Bedrock, EKS), GCP (Vertex AI, Cloud Run, GCS), Azure, Docker, Kubernetes, Git, CI/CD

Data: PostgreSQL, MySQL, MongoDB, PySpark, Apache Impala, pandas

EDUCATION

Rose-Hulman Institute of Technology

Aug. 2020 – May 2024

Bachelor of Science in Computer Science

Terre Haute, IN